

Engineering White Paper Roadmap

Version 1.0 Release Plan and Version 2.0 Expansion Plan

Building a Networked, WOPR-Inspired Information and Telemetry Console for the NABU Personal Computer

A Reproducible, Evidence-Driven Release Roadmap

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Roadmap Revision | August 25, 2026

As-Built Status Update | August 26, 2026

Original 25 Aug baseline (original label: Current baseline)	NCC-SA-260820-LV1
Original 25 Aug physical status (original label: Current physical status)	VERIFIED
Original 25 Aug Version 1.0 state	IN PROGRESS
Endurance operation	ACTIVE
Version 2.0	DEFERRED UNTIL V1.0 RELEASE
Current accepted baseline	NCC-SPLASH-260826-LV2
Current artifact size	000001.bin / 000001.NABU: 59,501 bytes each
Current SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
Current MAME / physical status	OWNER VERIFIED / OWNER VERIFIED
Current Version 1.0 state	FEATURE-COMPLETE RELEASE CANDIDATE; FINAL FREEZE / PUBLIC RELEASE PENDING

BUILD UPON ONLY. DO NOT STRIP DOWN.

Document Control

Field	Value
Document title	NABU Command Center: Engineering White Paper Roadmap
Document purpose	Version 1.0 release plan and Version 2.0 expansion plan
Roadmap revision	Original roadmap: August 25, 2026 As-built status update: August 26, 2026
Author	Derek Leger
Primary platform	NABU Personal Computer

Field	Value
Original 25 Aug verified baseline (original label: Current verified baseline)	NCC-SA-260820-LV1
Original 25 Aug artifact size	000001.bin and 000001.NABU: 51,421 bytes each
Original 25 Aug SHA-256	F8E440B8C6CD2E825914AEF1FD8E5B77B81926B52C0C6C539CCF5C33380C0AEF
Original 25 Aug physical NABU status	VERIFIED
Version 1.0 scope	Page 1 completion, Task Manager telemetry, streaming music, qualification, open-source release
Version 2.0 scope	Paging, Page 2 expansion, full-page Page 3 Task Manager / diagnostics
Current accepted baseline	NCC-SPLASH-260826-LV2
Current artifacts	000001.bin and 000001.NABU: 59,501 bytes each; byte-identical
Current SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
Current MAME status	OWNER VERIFIED
Current physical NABU status	OWNER VERIFIED
Current repository state	Owner-accepted LV2 release-candidate working tree; final LV2 commit/tag freeze not yet recorded
Current Version 1.0 state	Dynamic ZIP, all Page-1 live modules, Task Manager, Music, splash, integrated MAME regression, and physical-NABU regression complete; release freeze/endurance record/publication remain
Open-source license	MIT License — Copyright (c) 2026 Derek Leger; third-party material retains its own terms.

Evidence Boundary

This roadmap preserves the project evidence doctrine. VERIFIED is used only for behavior supported by direct build, artifact, MAME, or physical NABU evidence. ACTIVE identifies currently operating features or testing. PENDING and DEFERRED identify work that remains outside the verified release baseline. Version 1.0 is not frozen until the integrated qualification and Owner acceptance gates are complete.

Release Boundary

Version 1.0 will ship as the complete Page 1 Command Center with six primary modules, verified Task Manager telemetry, dynamic ZIP/location resolution, streaming music, endurance evidence, MAME regression, physical NABU regression, and the public open-source release. Pages 2 and 3 are reserved for Version 2.0.

AS-BUILT UPDATE — 26 AUG 2026: The feature work listed here has advanced materially. Dynamic ZIP/location, live Airspace/ADS-B, Task Manager truth corrections, production Music, splash, and integrated MAME plus physical-NABU regression are complete in the accepted LV2 candidate. Remaining release work is the quantified endurance record, final LV2 repository freeze/manifest, packaging, and public open-source publication.

Executive Summary

The NABU Command Center has reached a stable release-planning stage. The current protected baseline, NCC-SA-260820-LV1, has compiled, linked, generated byte-identical NABU artifacts, executed in MAME, executed on physical NABU hardware, and passed the Owner's physical full regression. Atomic Clock, Local Weather,

Earthquake, Space Weather, and Satellite / ISS are active. The remaining Version 1.0 work is focused rather than expansive: complete dynamic ZIP-to-city/state resolution, replace mock Airspace data with verified live ADS-B telemetry, audit and correct Task Manager telemetry, activate the production streaming-music feature, run the integrated regression and endurance gates, then freeze, document, and release the project as open source.

HISTORICAL SNAPSHOT NOTE — The Executive Summary paragraph above is retained verbatim as the 25 Aug planning snapshot. As of 26 Aug, ZIP resolution, Airspace, Task Manager corrections, Music, splash, integrated MAME regression, and physical-NABU regression have all advanced beyond that snapshot.

Version 2.0 is intentionally separated from the first public release. It will introduce dashboard paging, the six-module Page 2 expansion, and the dedicated full-page Page 3 Task Manager / diagnostics console. This keeps Version 1.0 centered on completing, qualifying, and publishing the already mature Page 1 system.

Release at a Glance

Area	Current state	Release meaning
Page 1 core	CURRENT 26 AUG: 6 of 6 release-scope primary services/modules implemented; integrated LV2 OWNER VERIFIED. ORIGINAL 25 AUG: 5 of 6 primary telemetry modules ACTIVE / VERIFIED	Current: Page-1 feature completion achieved; preserve and freeze exact accepted LV2. Original target: Complete Airspace, then qualify the full page
Task Manager	CURRENT 26 AUG: telemetry provenance audit/corrections complete; three internal pages verified. ORIGINAL 25 AUG: ACTIVE diagnostics; telemetry audit pending	Current: preserve truthful LIVE/CALC/MAP/CONST/UNAVAILABLE diagnostics. Original target: verify live fields and labels ORIGINAL 25 AUG WORDING: Verify live fields and label calculated/static fields accurately
Streaming Music	CURRENT 26 AUG: production bounded Music implemented and integrated; single stream, B toggle, keyboard C unbound. ORIGINAL 25 AUG: PENDING production activation	Current: accepted v1.0 implementation is cache-first AY-ready Music, not live audio streaming. Original target: required before Version 1.0 freeze ORIGINAL 25 AUG WORDING: Required before Version 1.0 freeze
Physical NABU	CURRENT 26 AUG: exact NCC-SPLASH-260826-LV2 integrated runtime OWNER VERIFIED on physical NABU. ORIGINAL 25 AUG: VERIFIED current baseline	Current: final integrated physical regression achieved; exact release freeze still pending. ORIGINAL 25 AUG WORDING: Repeat final regression on the Version 1.0 candidate
Endurance testing	CURRENT 26 AUG: ACTIVE; final quantified duration/interruption release record still pending. ORIGINAL 25 AUG: ACTIVE	Continue only evidence-based duration/recovery logging before final release claims. ORIGINAL 25 AUG WORDING: Continue and record duration/interruption evidence
Version 2.0	CURRENT 26 AUG: DEFERRED; no Page-2/Page-3/dashboard-paging implementation authorized before v1.0 freeze/release.	Paging, Page 2, and full-page Page 3 diagnostics remain post-v1.0 work. ORIGINAL 25 AUG WORDING: Paging, Page 2, and full-page Page 3 Task Manager after Version 1.0 release

Version 1.0 Immediate Order

- Dynamic ZIP -> city/state Gateway resolution
- Airspace / ADS-B real source verification and live integration

- Task Manager telemetry verification and correction
- Streaming Music production activation
- Full Version 1.0 integrated qualification
- Endurance qualification and Owner freeze
- Engineering White Paper / release documentation
- Public Git and open-source Version 1.0 release

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0. Purpose

This roadmap updates the previously established NABU Command Center engineering roadmap and changes the release boundary.

Version 1.0 will be a complete, qualified Page 1 release with six primary Command Center modules, working Task Manager telemetry, streaming music, dynamic ZIP/location resolution, full MAME and physical NABU regression, and an open-source public release.

Pages 2 and 3 are deferred to Version 2.0.

The standing engineering rule remains:

BUILD UPON ONLY. DO NOT STRIP DOWN.

Preserve verified functionality, graphics, controls, transport behavior, and accepted module presentation while completing Version 1.0.

1. Current Known-Good Baseline

Current verified build:

NCC-SA-260820-LV1

Artifacts:

000001.bin 51,421 bytes 000001.NABU 51,421 bytes

Artifact relationship:

Byte-identical

SHA-256:

F8E440B8C6CD2E825914AEF1FD8E5B77B81926B52C0C6C539CCF5C33380C0AEF

Verification status:

- Compilation: VERIFIED
- Linking: VERIFIED
- 000001.bin generated: VERIFIED
- 000001.NABU generated: VERIFIED
- MAME execution: VERIFIED
- Physical NABU execution: VERIFIED
- Owner physical full regression: VERIFIED
- Satellite / ISS live integration: VERIFIED
- Continuous physical-hardware endurance operation: ACTIVE

This remains the protected baseline until a stronger verified Version 1.0 candidate replaces it.

AS-BUILT UPDATE — 26 AUG 2026: A stronger accepted candidate now exists: NCC-SPLASH-260826-LV2; 000001.bin and 000001.NABU are 59,501 bytes each and byte-identical; SHA-256 5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087. Compilation/linking, MAME execution, Weather maximized-view correction, integrated modules, physical transfer/execution, and physical GUI/runtime are Owner verified. Repository freeze remains a separate pending gate.

2. Version 1.0 Product Scope

Version 1.0 is the complete Page 1 Command Center release.

Primary Page 1 modules:

1. Atomic Clock
2. Local Weather
3. Earthquake
4. Space Weather

5. Satellite / ISS

6. Airspace / ADS-B

Additional Version 1.0 capabilities:

- NWS weather alerts
- rolling weather trends
- dynamic ZIP/location selection
- Task Manager application/system telemetry
- streaming music
- global refresh behavior
- current graphical dashboard
- physical NABU operation
- continuous endurance testing
- documented Gateway and NABU build/deployment workflow
- public open-source release

Version 1.0 will NOT wait for Page 2 or Page 3.

AS-BUILT UPDATE — 26 AUG 2026: The release-scope count of six primary services includes the persistent Atomic Clock. In the final dashboard grid, five live information tiles plus NABU Task Manager are visible while the Atomic Clock remains a global header service. This terminology clarification does not change the v1.0 release scope.

3. Version 1.0 Module Status

1. ATOMIC CLOCK - ACTIVE / VERIFIED

Live date and time with the central Command Center clock display.

Required for Version 1.0:

Preserve current verified behavior.

2. LOCAL WEATHER - ACTIVE / VERIFIED

Current capabilities:

- live temperature
- humidity
- wind
- pressure
- current condition
- rolling graphical weather trends
- NWS weather alerts
- refresh behavior
- dashboard integration

Known remaining cleanup:

Dynamic ZIP to friendly city/state resolution.

Current arbitrary ZIP behavior is already proven to submit unlisted five-digit ZIP requests and retrieve weather data.

Current defect:

The NABU client falls back to UNRES for ZIPs outside the small set of built-in friendly profiles.

Version 1.0 target:

- user enters any supported five-digit ZIP
- NABU sends the bounded ZIP request
- Gateway resolves locality/city/state
- NABU receives a bounded returned location label
- client displays the resolved city/state
- no large ZIP database is stored in the NABU client
- existing weather behavior is preserved
- truthful fallback is shown when resolution fails

Example intended behavior:

85234 -> GILBERT AZ

AS-BUILT UPDATE — 26 AUG 2026: COMPLETED. Dynamic ZIP/location resolution is implemented Gateway-side with bounded NABU request/response records; the example 85234 -> GILBERT AZ is part of current production behavior. The final LV2 also includes the focused maximized-Weather renderer correction that removed the index/bounds stack-corruption defect found in LV1.

3. EARTHQUAKE - ACTIVE / VERIFIED

Current capabilities:

- live USGS data
- Local view
- Global view
- magnitude
- depth
- location/region
- age/event information
- target selection
- existing graphics

Version 1.0 target:

Preserve current verified behavior.

4. SPACE WEATHER - ACTIVE / VERIFIED

Current capabilities:

- NOAA / SWPC live data
- Kp

- solar wind
- magnetic conditions
- F10.7 where available
- historical trends
- storm threshold display
- readable graphical presentation
- live source/severity information

Version 1.0 target:

Preserve current verified behavior.

5. SATELLITE / ISS - ACTIVE / VERIFIED

Current capabilities:

- real ISS identity
- NORAD ID
- latitude
- longitude
- altitude
- velocity
- freshness/time information
- bounded SA01 records
- existing graphical/vector orbital presentation
- MAME verification
- physical NABU verification

Version 1.0 target:

Preserve current verified behavior.

6. AIRSPACE / ADS-B - PENDING

This is the final unfinished primary Page 1 telemetry module.

Version 1.0 goal:

Replace the current clearly identified mock aircraft display with real live aircraft telemetry.

Preferred source direction:

- local readsb / dump1090-class receiver source when available
- otherwise a separately verified provider after confirming availability,

authentication, terms, fields, update behavior, and rate limits

Planned bounded aircraft fields:

- callsign/identity where available
- position

- altitude
- ground speed
- heading/course
- target freshness
- selected target details
- source/health state

Presentation:

- preserve the existing graphical Airspace mockup
- replace mock tracks with real tracks
- keep a bounded target count suitable for the NABU
- Gateway filters and reduces aircraft data
- NABU does not ingest raw ADS-B JSON or large feeds
- no fabricated aircraft

Implementation sequence:

A. Verify real source B. Implement Gateway collector/cache C. Define bounded aircraft record D. Add focused Gateway tests E. Add NABU parser/cache/refresh F. Bind live data to existing Airspace renderer G. Add/verify refresh behavior H. Build I. Owner MAME acceptance J. Physical NABU acceptance K. Freeze

AS-BUILT UPDATE — 26 AUG 2026: COMPLETED / VERIFIED. Production Airspace uses ADSB.lol around the selected ZIP/location with a 100 NM query radius, reduces to at most three aircraft, and publishes bounded AS01 state. The accepted display is the owner-frozen 3D perspective/vector visualization, not a top-down radar redesign. Airspace LV3 passed Owner MAME acceptance and is included in the final physical-LV2 integrated regression.

4. Version 1.0 Task Manager

The existing Task Manager remains part of Version 1.0.

Current status:

ACTIVE as application/system diagnostics.

Current useful live/internal information includes:

- transport state
- sequence information
- bytes received
- active screen/module
- mode/state
- rendering counters
- full/dirty/static/dynamic draw counts
- current application state
- pause/sound state
- other internal counters

Current fixed/calculated information includes:

- platform specifications
- map/build-derived memory values
- calculated resource information

Version 1.0 goal:

Ensure the Task Manager's useful live fields are based on real application/system telemetry and are not merely decorative mock values.

Required classification:

LIVE = directly changing runtime/application state CALC = calculated value MAP = map/build-derived value CONST = fixed machine/platform specification

Version 1.0 Task Manager work:

- verify each displayed field's origin
- remove or relabel any mock values
- retain real runtime telemetry
- preserve useful fixed/calculated diagnostics with accurate labels
- verify counters change when expected
- verify transport values correspond to actual runtime activity
- verify active module/screen reporting
- verify refresh/failure state where instrumented
- keep the current bounded Page 1 Task Manager presentation
- do not implement the planned full-page Page 3 Task Manager in Version 1.0

Acceptance:

Task Manager must be demonstrably useful as a real application/system diagnostics module.

AS-BUILT UPDATE — 26 AUG 2026: COMPLETED / VERIFIED. Task Manager provenance and labels were corrected; the current truth classes are LIVE, CALC, MAP, CONST, and UNAVAILABLE. The module has three internal pages navigated with Left/Right; this is not the deferred Version-2 dashboard Page 3 console.

5. Version 1.0 Streaming Music

Streaming Music is now in Version 1.0 scope.

Status:

PENDING ACTIVATION / PRODUCTION IMPLEMENTATION

Existing placeholder/temporary music behavior is not the final Version 1.0 music feature.

Approved control direction:

B = toggle music C = cycle genre if final control audit confirms the key remains available R = refresh music status/state where required by final design

Planned genres:

01 Classical 02 Jazz / Standards 03 Blues / R&B 04 Country 05 Pop / Rock 06 Reggae / World 07 Electronic / Dance 08 Random / Mixed

Architecture:

Gateway:

- source acquisition
- caching
- failover
- conversion/reduction

NABU:

- bounded playback/control
- status display
- user controls

AY direction:

Channel A:

UI cues / sound effects where appropriate

Channels B/C:

music where appropriate

Required evidence before Version 1.0 freeze:

- exact playback architecture documented
- source/protocol verified
- NABU control path verified
- music does not regress UI sounds
- long-running playback does not destabilize transport/rendering
- MAME acceptance
- physical NABU acceptance

AS-BUILT UPDATE — 26 AUG 2026: IMPLEMENTED / VERIFIED. Final v1.0 Music is a cache-first bounded AY-ready service, not a live audio stream. The Gateway uses Mutoxia source material and Mido 1.3.3 conversion; the NABU loads a 512-byte MU01 resource in eight proven 64-byte Store reads and performs 60 Hz event playback. B toggles Music; keyboard C remains unbound; the planned genre list/failover remains historical design material. AY A remains cue-owned; Music uses B/C.

6. Version 1.0 Endurance Testing

Physical-hardware endurance testing continues in parallel with development.

Current practice:

The known-good Command Center build is run continuously on physical NABU hardware, including 24/7 operation as practical.

Version 1.0 endurance goals:

- no freeze during extended operation
- clock continues

- transport continues
- Gateway refresh continues
- module navigation remains responsive
- repeated R refresh remains stable
- no progressive graphical corruption
- no progressive state corruption
- no apparent memory degradation
- stale/offline states recover cleanly
- Gateway restart recovery works
- Internet Adapter restart/recovery works where tested
- streaming music does not destabilize the system
- aircraft refresh does not destabilize the system

Record actual duration and interruptions before making quantified uptime claims.

AS-BUILT UPDATE — 26 AUG 2026: This rule remains current. The integrated LV2 candidate is Owner verified in MAME and on physical NABU, but this roadmap does not upgrade endurance to COMPLETE without a retained duration/interruption/recovery record. Keep the final release claim evidence-based.

7. Version 1.0 Integrated Qualification

After ZIP resolution, Airspace, Task Manager telemetry verification, and Music are complete, run one complete Version 1.0 qualification.

The integrated regression must verify:

Boot / Identity

- correct Version 1.0 build ID
- correct binary/artifact hashes
- normal boot

Atomic Clock

- clock running
- date/time presentation correct for configured source

Local Weather

- live weather
- arbitrary ZIP entry
- dynamic city/state label
- trend history
- NWS alert presentation
- refresh

Earthquake

- Local

- Global
- target selection
- live data

Space Weather

- live data
- graph
- severity/source
- history

Satellite / Iss

- live position/data
- graphical orbit
- refresh

Airspace / ADS-B

- live aircraft
- target selection
- track graphics
- freshness
- refresh

Task Manager

- real runtime telemetry
- counters
- transport state
- correct LIVE/CALC/MAP/CONST labeling

Music

- toggle
- genre controls
- playback
- status
- coexistence with UI sound

Global Behavior

- global R refresh
- no Weather regression
- no module regression
- navigation
- return to dashboard

- sound
- input
- graphics
- bounded transport
- no freeze

Environments

- MAME regression
- physical NABU regression
- endurance evidence

AS-BUILT UPDATE — 26 AUG 2026: INTEGRATED RUNTIME QUALIFICATION ADVANCED. The exact NCC-SPLASH-260826-LV2 candidate has Owner-verified MAME execution and physical-NABU transfer/execution/GUI regression after the final Weather correction. The historical 'genre controls' bullet is superseded: shipped v1.0 has no keyboard genre-cycle control. Final quantified endurance evidence and repository freeze remain separate gates.

8. Version 1.0 Freeze

Version 1.0 is frozen only after Owner acceptance.

Freeze evidence should include:

- final build ID
- final source commit
- final Git tag
- 000001.bin size
- 000001.NABU size
- SHA-256
- byte-identity result
- Gateway version/state
- MAME acceptance record
- physical NABU acceptance record
- endurance-test record
- acceptance checklist
- known limitations

Do not advance the final release candidate after freeze unless a real defect requires a new versioned candidate.

AS-BUILT UPDATE — 26 AUG 2026: NOT YET FROZEN. The current exact accepted LV2 artifact is not represented by a final LV2 commit/tag in the reviewed repository state. The last explicitly tagged full application freeze is the pre-Music state: commit dd1bad5e6699fd0ddba01fddac384b7419270b22, tag ncc-command-center-pre-music-v1-mame-verified. Do not call LV2 repository-frozen until the Owner authorizes the final commit/tag/manifest.

9. Version 1.0 Open-Source Release Plan

MIT LICENSE UPDATE — 31 AUG 2026: Derek-owned NABU Command Center source and documentation are released under the MIT License, Copyright (c) 2026 Derek Leger. The license permits community use, modification, redistribution, and derivative development subject to preservation of the copyright and permission notice. Third-party components, tools, services, and referenced material remain under their own licenses or terms and are not relicensed by this project.

Development remains in controlled local Git until Version 1.0 is ready.

At release:

- create/prepare the public Git repository
- publish the frozen Version 1.0 source
- publish the Gateway source
- publish build scripts
- publish required configuration examples
- publish tested binary artifacts
- publish hashes
- publish release notes
- publish screenshots
- publish MAME verification status
- publish physical NABU verification status
- publish known limitations
- publish contributor guidance

Documentation package:

- Engineering White Paper
- README
- Version history
- build instructions
- z88dk setup
- Internet Adapter setup
- Gateway installation/start instructions
- record/protocol documentation
- module architecture
- API/source provenance
- acceptance tests
- troubleshooting
- release manifest
- contributor/module development guide

Open-source objective:

Provide a reproducible NABU Command Center framework that other developers can build, run, study, improve, and extend.

AS-BUILT UPDATE — 26 AUG 2026: Documentation has advanced: the consolidated Master Production Engineering White Paper v1.1 and the v1.0 User Manual & Operator Guide now exist. Public Git preparation/open-source publication is still pending and must use the final frozen source/artifacts, not the current unfrozen working tree.

10. Version 1.0 Immediate Programming Order

CURRENT PRIORITY ORDER:

1. Dynamic ZIP -> city/state Gateway resolution
2. Airspace / ADS-B real source verification
3. Airspace Gateway implementation
4. Airspace NABU live client integration
5. Airspace MAME acceptance
6. Airspace physical NABU acceptance
7. Task Manager telemetry audit
8. Task Manager live-field correction/improvement
9. Streaming Music production activation
10. Music MAME acceptance
11. Music physical NABU acceptance
12. Full integrated Version 1.0 regression
13. Extended endurance qualification
14. Version 1.0 freeze
15. Packaging/documentation
16. Engineering White Paper final Version 1.0 update
17. Public Git preparation
18. Open-source Version 1.0 release

AS-BUILT UPDATE — 26 AUG 2026: CURRENT ORDER STATUS: items 1-12 (ZIP through full integrated regression) are complete at the accepted LV2 level. Item 13 remains an evidence-record gate; item 14 final freeze is pending; packaging/documentation is active; the Master White Paper and User Manual are built; public Git preparation and open-source release remain pending. Do not add new Version-1 feature scope before the freeze.

11. Version 2.0 Scope

Version 2.0 is the expansion release.

It is intentionally deferred until Version 1.0 is complete, qualified, frozen, documented, and released.

AS-BUILT UPDATE — 26 AUG 2026: This deferral remains in force. The current three-page Task Manager is an internal module view within Version 1.0; it does not implement the planned Version-2 dashboard Page 3 diagnostics console.

Version 2.0 includes:

- dashboard paging infrastructure
- Page 2 six-module expansion
- Page 3 full-page Task Manager / diagnostics console
- integrated twelve-module qualification
- additional long-duration regression
- any larger post-1.0 enhancement work approved by the Owner

12. Version 2.0 Page 2

All Page 2 modules are deferred to Version 2.0.

1. Local Crime Watch

Planned:

- local incident type
- approximate area
- age
- severity/category
- bounded list
- compact graphical positioning

2. Radiation Detection

Planned:

- local radiation data when a verified local source/sensor exists
- global radiation data
- current level
- status
- trend
- freshness
- bounded history graph

3. Global Disaster Map

Planned:

- graphical world map
- event markers
- compact event feed
- storms
- floods
- fires
- earthquakes

- other verified major events

4. Volcanic Activity

Planned:

- global volcano activity
- alert level
- location
- event age
- recent activity
- compact graphical indicators

5. Stock Ticker

Defaults:

- Dow Jones
- S&P
- NASDAQ

Planned:

- manual ticker entry
- current value
- change
- direction
- bounded trend/history

6. Marine Tracking / AIS

Planned:

- live AIS vessel tracking
- bounded regional display
- vessel identity
- position
- course
- speed
- destination/status where available
- target selection
- freshness/source state

13. Version 2.0 Page 3

Page 3 becomes a dedicated full-screen Task Manager / diagnostics console.

Planned sections:

System Status

- build ID
- runtime/uptime where verified
- current page/module
- current mode
- pause
- sound

Network / Transport

- RetroNET state
- current/last resource
- sequence
- bytes
- refresh activity
- failures/retries
- recovery state

Module Health

- Clock
- Weather
- Alerts
- Earthquake
- Space Weather
- Satellite
- Airspace
- Page 2 modules

Possible states:

Live

Stale

Offline

Error

Not Loaded

Rendering / Ui

- full redraws
- dirty redraws

- frame/activity counters
- graphics state
- input state where useful

Memory / Resources

- live values only where genuinely measurable
- map values
- calculated values
- fixed platform values
- explicit source/category label for each

Event Log

Short bounded rolling event list:

- module refresh
- stale record
- transport retry
- recovery
- alert
- module failure/success

14. Version 2.0 Paging

Planned:

Page Down:

current page moves up, next page enters from bottom

Page Up:

current page moves down, previous page enters from top

Rules:

- one bounded transition animation
- static/state-driven page after transition
- preserve dashboard performance
- verify actual NABU Page Up / Page Down key codes before implementation
- retain consistent module layout where sensible

AS-BUILT UPDATE — 26 AUG 2026: Page Up/Page Down key codes remain unverified in the current NABU/z88dk project evidence. Left/Right are verified and used for Task Manager internal paging, but they must not silently substitute for the planned dashboard paging design without Owner approval.

15. Version 2.0 Qualification

After Page 2 and Page 3 are implemented:

- qualify all twelve information/telemetry modules together
- qualify Page 1
- qualify Page 2
- qualify Page 3
- qualify paging
- qualify Music
- qualify failure isolation
- qualify stale/offline behavior
- qualify Gateway recovery
- qualify Adapter recovery
- qualify long-running operation
- MAME full regression
- physical NABU full regression
- endurance qualification
- freeze Version 2.0
- package/document/release

16. Pi-Class Gateway Appliance

A dedicated Pi-class Gateway appliance remains a planned packaging/deployment direction.

It may be implemented after Version 1.0 or as part of the Version 2.0 era, subject to Owner priority.

Purpose:

- consolidate modern API/source collection
- normalize/cache bounded records
- simplify deployment
- support repeatable community installations
- reduce dependence on a full Windows development environment for end users

Exact hardware/software details remain evidence-driven and are not frozen yet.

17. Release Milestones

Version 1.0

Complete when:

- six Page 1 modules are real and qualified
- Task Manager telemetry is verified/useful
- arbitrary ZIP location resolution works
- streaming music is active
- MAME regression passes

- physical NABU regression passes
- endurance testing is acceptable
- source/artifacts/docs are frozen
- public open-source repository is released

Version 2.0

Complete when:

- paging is implemented
- Page 2 six-module expansion is complete
- Page 3 full Task Manager is complete
- twelve-module system passes integrated qualification
- Version 2.0 MAME and physical regressions pass
- Version 2.0 is documented and released

18. Current Development Rules

BUILD UPON ONLY.

DO NOT STRIP DOWN.

Version 1.0 has priority.

Do not begin Page 2 or Page 3 implementation until Version 1.0 is frozen and released unless the Owner explicitly changes priority.

Use real data.

No fake production telemetry.

Keep modern API complexity on the Gateway.

Keep NABU data bounded and deterministic.

Preserve Golden Transport.

Preserve global R behavior.

Preserve graphical presentation.

Treat MAME and physical NABU verification separately.

Continue physical endurance testing.

Keep Git local/private during active Version 1.0 development.

Publish the complete reproducible project at Version 1.0 release.

AS-BUILT UPDATE — 26 AUG 2026: CURRENT RESOURCE GATE: the accepted LV2 map places BSS_END at \$FC7A and initial SP at \$FF00, leaving a 646-byte static-to-stack address-space margin. The existing single-page H Help remains; a proposed dense multi-page Help/About view was intentionally stopped rather than consume this margin or invent Page Up/Page Down key codes. Do not strip verified production features to create room.

19. Current Next Action

Immediate programming objective:

Dynamic Zip / City-State Resolution

Then:

Airspace / ADS-B

Then:

Task Manager Telemetry Verification

Then:

Streaming Music

Then:

Full Version 1.0 Qualification And Release

AS-BUILT UPDATE — 26 AUG 2026: CURRENT NEXT ACTION — 26 AUG 2026: preserve the exact accepted LV2 candidate; record final endurance evidence; complete release manifest/provenance; Owner-authorize and perform the final LV2 commit/tag freeze; package the exact artifacts; prepare public Git/open-source Version 1.0 release. No new v1.0 feature work is required before those release gates.

Appendix A. Version 1.0 Release Gate Matrix — Original 25 Aug Plan with 26 Aug Current-State Overlay

This matrix condenses the release conditions already defined in the roadmap into a single acceptance view.

Gate	Current state	Required result	Evidence at freeze
Dynamic ZIP resolution	CURRENT 26 AUG: COMPLETE / integrated accepted. ORIGINAL 25 AUG: PENDING cleanup	Arbitrary supported ZIP returns bounded city/state label	Gateway/client test + MAME/physical acceptance
Airspace / ADS-B	CURRENT 26 AUG: COMPLETE / live ADSB.lol / MAME + physical integrated accepted. ORIGINAL 25 AUG: PENDING	Real verified aircraft telemetry replaces mock tracks	Source provenance + tests + MAME + physical NABU
Task Manager	CURRENT 26 AUG: COMPLETE / truth audit + LV2 correction accepted. ORIGINAL 25 AUG: ACTIVE / audit pending	Live fields proven; CALC/MAP/CONST values labeled accurately	Field provenance audit + runtime observations
Streaming Music	CURRENT 26 AUG: COMPLETE / bounded single-stream Music integrated; B toggle; C unbound. ORIGINAL 25 AUG: PENDING	Production music control/playback stable with UI sound	Source/protocol evidence + MAME + physical NABU
Integrated Page 1 regression	CURRENT 26 AUG: OWNER VERIFIED in MAME and physical LV2. ORIGINAL 25 AUG: PENDING final	All six modules, refresh, navigation, graphics, sound, input stable	Owner acceptance checklist
Endurance	CURRENT 26 AUG: ACTIVE; quantified release record still pending. ORIGINAL 25 AUG: ACTIVE	Acceptable long-duration stability with recorded evidence	Duration/interruption log

Gate	Current state	Required result	Evidence at freeze
Release freeze	CURRENT 26 AUG: PENDING final LV2 commit/tag/manifest.	Final build ID, commit, tag, sizes, hash, docs	Release manifest
Open-source publication	CURRENT 26 AUG: PENDING; documentation substantially complete, public release not yet performed.	Public reproducible source + Gateway + docs + artifacts	Public repository/release

Immediate programming objective: DYNAMIC ZIP / CITY-STATE RESOLUTION -> AIRSPACE / ADS-B -> TASK MANAGER TELEMETRY VERIFICATION -> STREAMING MUSIC -> FULL VERSION 1.0 QUALIFICATION AND RELEASE.

Appendix B. 26 August 2026 As-Built Status

This appendix is the current-status overlay for the preserved 25 August roadmap. The original roadmap text remains in this document as the planning record; this appendix states which gates were actually completed and which release tasks still remain.

Gate / subsystem	26 Aug state	As-built note
Dynamic ZIP / city-state	COMPLETE / VERIFIED in integrated candidate	Gateway-side bounded location resolution; no large NABU ZIP database.
Atomic Clock	VERIFIED / PRESERVE	Current global header service retained.
Local Weather + Alerts + Trends	VERIFIED / PRESERVE	Final LV2 includes the maximized-Weather stack/bounds correction.
Earthquake	VERIFIED / PRESERVE	Local/Global live USGS data and target selection retained.
Space Weather	VERIFIED / PRESERVE	NOAA/SWPC live data and readable trends retained.
Satellite / ISS	VERIFIED / PRESERVE	Live ISS / NORAD 25544 vector presentation retained.
Airspace / ADS-B	COMPLETE / VERIFIED	ADSB.lol, 100 NM query, up to 3 retained aircraft, owner-frozen 3D perspective display.
Task Manager	COMPLETE / VERIFIED	Three internal pages; LIVE/CALC/MAP/CONST/UNAVAILABLE truth model.
Music	COMPLETE / VERIFIED	Mutopia/Mido host conversion; bounded MU01; B toggle; keyboard C unbound; AY B/C playback.
Splash	COMPLETE / VERIFIED	Version 1.0 splash, creator credit, fixed embedded march, any-key bypass.
Integrated MAME regression	OWNER VERIFIED	Exact NCC-SPLASH-260826-LV2 accepted.
Integrated physical NABU regression	OWNER VERIFIED	Exact LV2 transfer/execution/GUI operation accepted.
Endurance release record	ACTIVE / NOT YET FINALIZED	Do not publish quantified uptime without retained duration/interruption evidence.
Final LV2 Git freeze	PENDING	Current accepted LV2 bytes are not yet represented by final LV2 commit/tag.

Gate / subsystem	26 Aug state	As-built note
Open-source publication	PENDING	Package only after final freeze/manifest.

Current Accepted Candidate

Field	Value
BUILD_ID	NCC-SPLASH-260826-LV2
000001.bin	59,501 bytes
000001.NABU	59,501 bytes
Artifact relationship	Byte-identical
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
MAME	OWNER VERIFIED
Physical NABU	OWNER VERIFIED
Repository freeze	PENDING final LV2 commit/tag
Memory/resource note	BSS_END \$FC7A; initial SP \$FF00; 646-byte static-to-stack address-space margin

Physical Acceptance Evidence

Physical-NABU photographs have been removed from this public-facing edition for privacy. The MAME and physical-NABU verification statuses, build identity, hashes, and acceptance record remain unchanged.

Appendix C. Remaining Version 1.0 Release Work

Order	Release action	Acceptance rule
1	Preserve exact accepted LV2 bytes	Do not rebuild or edit simply to 'clean up' before freeze unless a concrete defect requires a new versioned candidate.
2	Record final endurance evidence	Retain actual duration, interruptions, restart/recovery observations, and exact build identity.
3	Complete release manifest	BUILD_ID, full artifact hashes, sizes, byte identity, Gateway state/dependencies, MAME status, physical status, known limitations.
4	Freeze repository state	Owner-authorized commit and annotated tag for the exact accepted LV2 source/artifacts/evidence.
5	Package documentation	Master Engineering White Paper v1.1, User Manual & Operator Guide, build/deployment instructions, protocol/provenance, acceptance tests.
6	Prepare public Git/open-source release	Publish the frozen source/Gateway/build scripts/config examples/artifacts/hashes/docs; no remote push before Owner authorization.

No-New-Scope Rule Before Freeze

BUILD UPON ONLY. DO NOT STRIP DOWN. Version 1.0 feature implementation is now complete at the accepted LV2 level. The release path is evidence, freeze, packaging, and publication—not new Page-2/Page-3 features or speculative memory recovery.

Version 2.0 Boundary — Preserved

- Dashboard paging infrastructure remains deferred until Version 1.0 is frozen and released.
- Page 2 six-module expansion remains deferred.
- The planned full-dashboard Page 3 diagnostics console remains deferred; the current v1.0 Task Manager's internal pages 1/3, 2/3, and 3/3 do not constitute that Version-2 page.
- Page Up/Page Down key codes remain unverified in current project evidence and must be verified before the planned paging implementation.
- The Pi-class Gateway appliance remains a future packaging/deployment direction.

Documentation Status

Document	26 Aug status
NABU Command Center Master Production Engineering White Paper v1.1	BUILT / corrected after independent Codex source/repository audit
NABU Command Center v1.0 User Manual & Operator Guide	BUILT
Original 25 Aug Roadmap	PRESERVED in this updated document with current-state overlays
Final release manifest / public repository release notes	PENDING final freeze